

LABORATORY OBSERVATION

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 1(1)

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1. TITLE

Create a webpage titled "College Information Portal"

2. AIM

To design and develop a **College Information Portal** webpage using HTML5 elements to display college information, student details, registration forms, images, audio and video content.

3. OBJECTIVE

The objectives of this experiment are:

- ☐ ☐ To understand the basic structure of an HTML5 webpage.
 - ☐ To create a college information portal using HTML components.
 - ☐ To use headings, paragraphs, lists and hyperlinks to organize information.
 - ☐ To create and display student information using HTML tables.
 - ☐ To create a student registration form using different HTML form controls.
 - ☐ To use text fields, email, password, radio buttons, checkboxes and select options.
 - ☐ To use HTML5 input types such as color, date, number and range.
 - ☐ To embed images, audio and video into a webpage.
 - ☐ To create navigation links to different sections of the webpage.
 - ☐ To understand the use of semantic elements such as <header>, <nav>, <section>, <article> and <footer>.
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4. THEORY

HTML (HyperText Markup Language) is used to create and structure content on web pages. HTML5 provides various elements for organizing information and supporting multimedia content.

In this experiment, a **College Information Portal** is created to display information about a college and its students. The webpage is divided into different sections such as About, Student Information, Registration and Media.

The <header> element is used to define the heading section of the webpage, while <nav> provides navigation links. The <section> and <article> elements are used to organize related content. Student information is represented using an HTML <table> with rows and columns. HTML forms provide different controls for collecting user information. In this experiment, text input, email, password, radio buttons, checkboxes, drop-down lists, textarea, color picker, date picker, number input and range input are used.

HTML5 also provides multimedia elements such as <audio> and <video> for embedding media files directly into a webpage. Images are embedded using the element, while hyperlinks are created using the <a> element.

5. ALGORITHM / PROCEDURE

- ☐ ☐ Create a new HTML file and declare the HTML5 document using <!DOCTYPE html>.
- ☐ Create the <head> section with character encoding, viewport and webpage title.
- ☐ Create a header containing the title **Student and College Information Portal**.
- ☐ Create a navigation bar with links to About, Student, Register and Media sections.
- ☐ Create the About section using <section> and <article>.
- ☐ Add information about the college using headings and paragraphs.
- ☐ Display different departments using an ordered list.
- ☐ Display college facilities using an unordered list.
- ☐ Add reference links to the college website and Google.
- ☐ Create the Student Information section.
- ☐ Create a table containing roll number, student name, branch and email.
- ☐ Add college logo and college building images using the element.
- ☐ Create the Student Registration section.
- ☐ Create a registration form using <form> and <fieldset>.
- ☐ Add input fields for name, email and password.
- ☐ Add radio buttons for selecting gender.
- ☐ Add checkboxes for selecting hobbies.
- ☐ Add a drop-down list for selecting the branch.
- ☐ Add a textarea for entering comments.
- ☐ Add color, date, number and range input controls.
- ☐ Add Submit and Reset buttons.
- ☐ Create the Media section.
- ☐ Embed an audio file using the <audio> element.
- ☐ Embed a video file using the <video> element.
- ☐ Create a footer containing copyright information.

- ❑ Save the HTML file and open it in a web browser.
 - ❑ Verify the navigation, table, registration form, images and multimedia elements.
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6. PROGRAM

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>College Information Portal</title>
</head>

<body>
  <center>
    <!--Heading Section-->
    <header>
      <h1>Student and College Information Portal</h1>
      <p>welcome to anits college portal</p>
    </header>
    <hr>
    <!--Nav Bar-->
    <nav>
      <a href="#about">About</a> |
      <a href="#students">Student</a> |
      <a href="#reg">Register</a> |
      <a href="#media">Media</a>
    </nav>
    <hr>
    <!--Part A-->
    <section id="about">
      <article>
        <h1>Welcome To ANITS</h1><br>
        <p>We are proud to be anitians</p><br>
        <p><b>Enter a realm where creativity ignites excellence at ANITS</b><br>
          <i>Home to the illustrious NAAC "A+" honour</i><br>
          <u>we're a cradle of greatness recognised by AICTE and Andhra
University.</u><br>
          2x<sub>2</sub>+x<sup>3</sup><br>
        </p>
        <h1>Departments</h1>
      </article>
    </section>
  </center>
</body>
</html>
```

```
<p>
<ol>
  <li>Chemical</li>
  <li>Computer Science</li>
  <li>Ai & MI</li>
  <li>DS</li>
  <li>Civil</li>
  <li>Mechanical</li>
  <li>ECE</li>
  <li>EEE</li>
  <li>IT</li>
</ol>
</p>
<h1>Facilities</h1>
<p>
<ul>
  <li>Library</li>
  <li>Play Ground</li>
  <li>Labs</li>
  <li>Canteen</li>
</ul>
</p>
<h1>Reference Links</h1>
<p>College Link</p>
<a href="https://anits.org/" target="_blank">Anits Website</a>
<p>Google Link</p>
<a href="https://www.google.com" target="_blank">Google</a>
</article>
</section>
<hr>
<!--Part B-->
<section id="students">
  <article>
    <h1>Student Info</h1>
    <table cellpadding="10" border="5">
      <tr>
        <th>Roll No</th>
        <th>Name</th>
        <th>Branch</th>
        <th>Email</th>
      </tr>
      <tr>
        <td>116</td>
        <td>JAy</td>
```

```

        <td>CSM</td>
        <td>JAy@gmail.com</td>
    </tr>
    <tr>
        <td>111</td>
        <td>Sandy</td>
        <td>ECE</td>
        <td>Sandy@gmail.com</td>
    </tr>
    <tr>
        <td>101</td>
        <td>Rahul</td>
        <td>CSE</td>
        <td>rahul@gmail.com</td>
    </tr>
</table>
</article>
</section>
<hr>
<section>
    
    
</section>
<hr>
<!--Part C-->
<section id="reg">
    <article>
        <form action="" method="">
            <fieldset>
                <legend>
                    <h1>Student Registration</h1>
                </legend>
                <p>
                    <label for="name">Name</label>
                    <input type="text" name="" id="name">
                    <br><br>
                    <label for="email">Email</label>
                    <input type="email" name="" id="email">
                    <br><br>
                    <label for="password">Password</label>
                    <input type="password" name="" id="password">
                    <br><br>
                    <label for="gender">Gender</label>
                    <input type="radio" name="gender" id="gender">Male

```

```
<input type="radio" name="gender" id="gender">Female
<input type="radio" name="gender" id="gender">Others
<br><br>
<label for="hobbies">Hobbies</label>
<input type="checkbox" name="" id="hobbies">Music
<input type="checkbox" name="" id="hobbies">Dance
<input type="checkbox" name="" id="hobbies">Reading
<br><br>
<label for="branch">Branch</label>
<select name="" id="branch">
  <option value="">Csm</option>
  <option value="">Cse</option>
  <option value="">Eee</option>
  <option value="">Ece</option>
  <option value="">Civ</option>
  <option value="">Mech</option>
</select>
<br><br>
<label for="comments">Comments</label>
<textarea name="" id="comments" rows="10" cols="50"
  placeholder="write your comments here"></textarea>
<br><br>
<label for="color"> Fav Color</label>
<input type="color" name="" id="color">
<br><br>
<label for="date">Date Of Birth</label>
<input type="date" name="" id="date">
<br><br>
<label for="number">Mobile Number</label>
<input type="number" name="" id="number">
<br><br>
<label for="range">Age</label>
<input type="range" name="" id="range">
<br><br>

<button type="submit">Submit</button>
<button type="reset">Reset</button>
</p>
</fieldset>
</form>
</article>
</section>
<!--Part D-->
<section id="media">
```

```

<article>
  <h1>Audio</h1>
  <audio controls>
    <source src="clg_audio.mp3" type="audio/mpeg">
  </audio>
  <h1>Video</h1>
  <video width="100" controls>
    <source src="clg_vide0.mp4" type="video/mp4">
  </video>
</article>
</section>
<hr>
<footer>
  <p>&copy; 2026 Anits College Info Portal. All Rights Reserved.</p>
</footer>
</center>
</body>

</html>

```

7. CODE EXPLANATION

<!DOCTYPE html>

Declares that the document uses HTML5.

<header>

Defines the heading section containing the title and introductory information.

<nav>

Creates the navigation bar containing links to different sections of the webpage.

Creates a hyperlink that navigates to the About section of the same webpage.

<section>

Groups related content into separate sections such as About, Students, Registration and Media.

<article>

Represents an independent block of content within a section.

<h1>

Defines major headings for the webpage and its sections.

<p>

Defines paragraphs of text.

Creates an ordered list for displaying the departments.

Creates an unordered list for displaying college facilities.

Defines individual items in ordered and unordered lists.

Creates external hyperlinks such as the college website and Google.

<table>

Creates a table for displaying student information.

<tr>

Defines a row in the table.

<th>

Defines a table heading.

<td>

Defines a table data cell.

Embeds images such as the college logo and college building.

<form>

Creates the student registration form.

<fieldset>

Groups related form controls together.

<legend>

Provides a title for the fieldset.

<input type="text">

Accepts text such as the student's name.

<input type="email">

Accepts an email address.

<input type="password">

Creates a password input field.

<input type="radio">

Allows the user to select one option from a group, such as gender.

<input type="checkbox">

Allows the user to select multiple options, such as hobbies.

<select>

Creates a drop-down list for selecting a branch.

<textarea>

Provides a multi-line field for entering comments.

<input type="color">

Provides a color selection control.

<input type="date">

Provides a date selection control.

<input type="number">

Allows the user to enter a numerical value such as a mobile number.

<input type="range">

Creates a slider for selecting a value within a range.

<button type="submit">

Submits the registration form.

<button type="reset">

Resets the form fields to their initial values.

<audio controls>

Embeds an audio file and provides playback controls.

<video controls>

Embeds a video file and provides playback controls.

<footer>

Defines the bottom section of the webpage containing copyright information.

8. TEST CASES AND EXECUTION DATA

I would not just write three random test cases. I would actually execute them and record the results.

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Open the HTML file in a browser	College Information Portal should be displayed	Webpage displayed successfully	Pass
TC-02	Click About navigation link	About section should be displayed	About section displayed	Pass
TC-03	Click Student navigation link	Student information section should be displayed	Student section displayed	Pass
TC-04	Click Register navigation link	Registration form should be displayed	Registration form displayed	Pass
TC-05	Click Media navigation link	Audio and video section should be displayed	Media section displayed	Pass

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-06	Verify student table	Roll number, name, branch and email should be displayed	Student details displayed	Pass
TC-07	Enter name, email and password	Input fields should accept the values	Values accepted	Pass
TC-08	Select gender and hobbies	Selected options should be accepted	Options selected successfully	Pass
TC-09	Select a branch from the drop-down list	Selected branch should be displayed	Branch selected successfully	Pass
TC-10	Enter comments in textarea	Comments should be accepted	Comments entered successfully	Pass
TC-11	Select date, color, number and range values	Corresponding controls should accept values	Values accepted	Pass
TC-12	Click Reset	Form fields should be cleared/reset	Form reset successfully	Pass
TC-13	Click college website link	College website should open in a new tab	Link opened successfully	Pass
TC-14	Play the audio and video	Multimedia files should play with controls	Media played successfully	Pass

9. OUTPUT

1.

2.

3.

4.

5.

6.

7.

8.

9.

Student and College Information Portal

welcome to anits college portal

[About](#) | [Student](#) | [Register](#) | [Media](#)

Welcome To ANITS

We are proud to be anitians

Enter a realm where creativity ignites excellence at ANITS
Home to the illustrious NMAC "A++" honour
we're a cradle of greatness recognised by AICTE and Andhra University.
2x₂1x²

Departments

Chemical
Computer Science
AI & ML
DS
Civil
Mechanical
ECE
EEE
IT

Facilities

Library
Play Ground
Labs
Canteen

Reference Links

College Link
[Anits Website](#)
Google Link
[Google](#)

Student Info

Roll No	Name	Branch	Email
116	Jay	CSM	Jay@gmail.com
111	Sandy	ECE	Sandy@gmail.com
101	Rahul	CSE	rahul@gmail.com



Student Registration

Name

Email

Password

Gender ☐ Male ☐ Female ☐ Others

Hobbies ☐ Music ☐ Dance ☐ Reading

Branch

Csm

write your comments here

Comments

Fav Color

Date Of Birth

mm/dd/yyyy

Mobile Number

Age

Submit

Reset

Audio

0:00 / 0:00

Video

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10. RESULT

Thus, the **College Information Portal** webpage was successfully designed using HTML5. The webpage successfully includes college information, navigation links, department and facility lists, student information tables, a student registration form, images, audio, video and external hyperlinks. Various HTML5 form controls and multimedia elements were also successfully implemented and tested.